

ARWEN BOROWIAK

Data Scientist / Data & AI Engineer | German Citizen (EU/EFTA)

PROFESSIONAL SUMMARY

Data scientist and engineer with ~3 years of experience applying statistical modeling, machine learning (scikit-learn), and cloud data architecture (Databricks/Snowflake) to solve operational and forecasting problems in pharmaceutical and consulting environments. M.S. in Business Analytics (GPA 4.0, Summa Cum Laude, Penn State Smeal). Track record designing automated, self-correcting data pipelines and statistical models that scaled to enterprise-wide standards — with a strong foundation for extending into agentic AI system design. EU citizen, eligible for Swiss work sponsorship without restriction.

SKILLS & TECHNOLOGIES

AI/ML & Analytics: Python (scikit-learn, pandas, NumPy) · Statistical & predictive modeling · ARIMA forecasting · Feature engineering · Minitab · R

Data Engineering & Cloud: Databricks · Snowflake · SQL · ETL pipeline design · Teradata · Alteryx · Power Automate

Systems & BI: SAP (P2R/H2R) · Kinaxis · E2Open/TMS · PowerBI · Tableau

Languages: German (native) · English (fluent) · French (conversational)

PROFESSIONAL EXPERIENCE

Digital Supply Chain Excellence Associate — Bayer

Sep 2025 – Present

- Designed and deployed Python/SQL data pipelines with statistical validation logic across multi-source inventory and demand datasets, cutting data latency 40%.
- Led migration of 15+ legacy SAP (P2R/H2R) workflows to a Databricks-Snowflake cloud architecture, enabling scalable model-ready analytics.
- Built a Python calendar-normalization algorithm resolving 52/53-week ISO discrepancies, adopted as the enterprise-wide standard — an example of designing a self-contained rule-based engine now used autonomously across teams.

Demand Planning & Distribution Analyst — Johnson & Johnson

Sep 2024 – Aug 2025

- Built and maintained large-scale datasets driving weekly order-release and inventory-planning decisions for 20+ distributors.
- Designed a statistical KPI model for order value/volume thresholds, reducing distributor stock-imbalance risk ~18%.
- Developed a Python-based real-time compliance monitoring model, cutting manual audit time 25%.

Supply Chain Operations Intern — Bayer

May – Aug 2024

- Orchestrated \$10M+ in consumer health logistics via E2Open/TMS, achieving 98% OTIF through data-driven load planning.
- Consolidated 5 departmental BI reports into a unified PowerBI/Tableau dashboard for 30+ team members.
- Automated daily workflows with Python and Power Automate, generating \$50K/month in savings.

Supply Chain Transformation Consultant — Genpact

May – Aug 2023

- Designed data models and integrated Kinaxis Supply Planning APIs for a \$77B pharma transformation programme.
- Supported Kinaxis supply-planning integration and production-scheduling model design.

Quality Control & Continuous Improvement Associate — GSK

May – Aug 2022

- Applied Lean Six Sigma/5S methodology to standardize 20 analytical lab workstations.
- Built a real-time KPI tracking database for supervisor visibility.

EDUCATION

M.S. in Business Analytics — Summa Cum Laude, 4.0 GPA — Penn State, Smeal College of Business *Aug 2025*

B.S. in Supply Chain & Information Systems — Pennsylvania State University *May 2024*

CERTIFICATIONS

Google Data Analytics Certificate · Google Business Intelligence Certificate · Lean Six Sigma Yellow Belt · APICS CSCP (in progress) · APSM – ISM (in progress)

LEADERSHIP & ATHLETICS

NCAA Division I Fencer & Team Captain, Penn State (2020–2025) — FIE World Ranking #2 (Foil), 4x NCAA All-American, raised ~\$300K in alumni donations as captain.